

Examiner reports

Q3.

There was a full spread of marks seen in this question with about a quarter of the students scoring 8 or more of the 10 marks available, but a similar proportion scoring 2 or less.

- (a) Most students recognised that this required integration by parts, although some just ‘integrated’ the t term and the cosine term independently, for no credit. Some tried to use substitution, and indeed some students substituted $u = \frac{\pi t}{4}$ and then used parts. Most identified the parts the right way round but those who did not should have found the integral becoming impossibly harder but they were often determined to get a result, again for no credit. Of those who started correctly, many made a mistake in writing down the parts formula, often confusing the sine and cosine terms. Many had the required coefficient $\frac{4}{\pi}$ inverted. Many who had the coefficients correct at this stage then made an error when integrating the sine term, often getting $\frac{8}{\pi}$ or similar. Many made a mistake when simplifying a correct expression, which was not penalised until part (b).
- (b) Most students separated the variables correctly, and usually used the correct notation. Occasionally the x was dropped or the dx or dt did not appear or was in the wrong place. A few attempted to ‘integrate’ the expression as given in the question to produce answers of no value. Most gained the mark for integrating x correctly, although some used logarithms due to incorrect separation. Most of the students also gained the method mark for using their result from part (a) in an attempt to find a constant of integration. Some students just substituted $t = 0$, $x = 4$ into the differential equation itself, which made no sense. Those who had the correct form of the integral from part (a) could then score the remaining method mark for calculating firstly x^2 and then x . Many were correct to this point and then misunderstood the request to give their answer to the nearest centimetre or they evaluated using degrees rather than radians. About 8% of the entry gave a fully correct solution.

Q4.

For part (a), some student wrote down the correct differential equation apparently fully understanding all the information given and interpreting it correctly. However, all sorts of errors abounded in other attempts, some not even involving a derivative, and some with derivatives in x and y . Some attempts bore little or no relation to the information given, although credit was given if $\pm (2 - h)$ appeared in an otherwise incorrect expression. Many had a spurious t and l or h , either as a multiple or power, and the k appeared in a variety of places. Some students did not even form an equation, leaving a proportionality sign in their answer.

In part (b), most students knew they were expected to separate the variables and did it correctly, although there were some notation errors in the positioning of dx , at the front rather than the rear of the integrand. Those who failed to separate the variables, just produced nonsense.

Most students integrated $k dt$ successfully, but relatively few made good progress with the integral in x . No guidance was given, but students split fairly evenly between attempts by

parts and by substitution, with similar success, or lack of it, via either method. Many students just integrated each term in the product; presumably this was an attempt at parts. Those who showed the parts and how the integral might be achieved scored a method mark, but many could not integrate $\sqrt{(2x-1)}$ correctly, making an error in the required coefficient.

Those who used substitution usually chose to put $u = 2x - 1$ but did not always change their whole integral to be in terms of u ; a missing du was penalised for the method mark. Some with a correct first stage then did not multiply out to obtain an integrable form, but just integrated each term in the product. All students, no matter what their attempt at the integral, could obtain a method mark if they included a constant and tried to find it using the given initial conditions. Those few who did integrate and find the constant correctly, usually went on to calculate the final requested value of t correctly.

Q5.

A full range of marks were seen in this question. Some candidates had little or no idea as to how to solve a differential equation, whereas some gave concise and correct solutions and could use their solution to solve the problem in the given context.

In part (a)(i) most candidates separated the variables correctly, although some poor notation was seen and some mishandled the $\sqrt{x}$. Many also made errors in the integration, often getting the coefficient wrong in one or both integrals. Although many did add an arbitrary constant, many also failed to complete the question as requested and by writing x in terms of t . Of those that did, many failed to square their expression in t correctly; $(f(t) + C)^2 = (f(t))^2 + C^2$ was a common error. Many apparently didn't review their solution when given the form of answer in part (a)(ii).

When manipulating an expression containing a constant some, for example, renamed $\frac{C}{2}$ as C

and sometimes then confused themselves; whereas others correctly introduced a new letter for the revised constant.

In (a)(ii) many went on to find a correct value for the constant and give the required form, as they used a previous line of working from part (a)(i) rather than their final answer. Any attempt to find a constant using the information given was credited, but many tried to force their solution into the form of the answer given, rather than review their work for an error.

In part (b)(i) the question confused a lot of candidates and many assumed the greatest height occurred at $\cos bt = 0$ and not -1 . Many candidates didn't attempt this part.

In (b)(ii) for those who set up a correct equation, the most common error here was to use degrees instead of radians; so the more common answer was 207.3, it apparently not occurring to candidates that a time of over 3 minutes was rather long for a fairground car to reach a height of 5 m. Many did not set up a correct form of the equation and so could gain no marks.

Q6.

In part (a) most candidates at least used $\frac{dA}{dt}$ in their attempt at a differential equation

although

some did use other letters. Very few equated to a constant of either sign. The term “constant rate” as given in the question just wasn’t understood, with most taking the rate of change to be proportional to the surface area, or to time or functions of these such as an exponential function. Only about 5% of the candidates set up the correct equation, which meant they could achieve little in part (b)(i).

In part (b)(i) those who had set up a correct equation in either $\pm k$ usually completed correctly, using integration and the conditions given. Some did attempt to integrate their differential equation, and this was given credit; many did get as far as a value for k , in which algebraic and numerical errors were condoned for the method mark, but somewhat less than 15% of the candidates did get this far.

In (b)(ii) most candidates were able to interpret the given solution in part (b)(i) to answer this correctly with over 60% of the candidates gaining this 1 mark.

Q7.

- (a) Attempts at this question varied from no attempt, through superficial attempts to some fully correct solutions. Most could separate the variables correctly, although a few did attempt “integrals” of $f(x)dt$. Some candidates inverted the given differential equation and so could attempt direct integration, which was a quite acceptable method. Most integrated kdt correctly, but the integral of $(x+1)^{-\frac{1}{2}}$ defeated many. Often there was an error in the power, before or after integration, and also in the coefficient. However, most candidates did include a constant and attempted to find it from the given initial conditions, for which they gained credit. However, some did some algebraic manipulation before adding a constant, and so could gain no credit, as this was then incorrect. Many candidates who had correctly integrated the expressions failed to gain the last mark, which was for giving the answer in the form $x = f(t)$, because they couldn’t square their expression correctly, often omitting the “middle” term.
- (b) Candidates who had interpreted the context correctly could gain at least one mark here by substituting $t = 60$ into their result from part (a) and solving for x . However, many confused the given $t = 60$ with the given initial condition $x = 80$. Relatively few correct answers were seen, with many candidates obtaining results of over 100% or indeed negative results. These were not apparently queried, with little evidence of checking for errors.
- (c) (i) Very few candidates wrote down a correct differential equation. Some who had seemingly understood the context, wrote down the correct expression in A , but did not include the constant of proportionality. Many candidates mixed up the variables with x , and less so y , appearing, although most had a “differential equation” of some sort. Some candidates even included units in their differential equation, which was condoned if at the end of a correct expression.
- (ii) For many candidates this was the only part of this question, they attempted. Candidates who had understood the context and started with $A = 4.5$ could mostly solve the equation for t successfully. The relatively rare error was in the algebraic manipulation or in attempts to take logarithms, such as $\log(1 + 4e^{-0.09t}) = \log 1 - 0.09 \log 4t$ or similar. Some made an error in miscopying -0.09 .

Those candidates who commonly started with $A = 50$ or $A = 0.5$ encountered the need to take the logarithm of a negative number in their working or obtained a negative value for t , which most then chose to ignore. They gained no credit. Some did realise their error in interpreting the context, and reverted to the correct value for A .

Q8.

The full range of marks was seen in this question. Some candidates showed no understanding of how to proceed to solve the differential equation or how to use the given result to answer part (b). In contrast, some gave a concise and accurate derivation of the solution and understood how to use it in part (b). In part (a), algebraic error in the exponential term was common in attempted rearrangements, although some candidates did not separate x and t before attempting integration.

Similarly, in the integration of the exponential term, the sign or coefficient was often incorrect, although most candidates could integrate kt correctly. In part (a)(ii), most candidates had included a constant, and of these many changed its value whilst rearranging, by dividing by 2 or changing signs without saying so. Due to sign errors, many could not take the logarithm of both sides correctly; although taking logarithms of a negative number was condoned for the method mark, $\ln(-kt^2)$ becoming $-\ln(kt^2)$ or similar was not. There was a lot of fudging of signs and coefficients in attempts to get to the given answer.

In part (b), candidates who went awry did not seem to realise they had been given the solution for x , and that the given 6000 matched the initial conditions given in part (a).

Many seemed to think that the 6000 was $\frac{dx}{dt}$, and that the given 10 was t , and they solved for x . Those who did write down the correct expression usually evaluated it correctly, with a few overlooking the request for the nearest hundred, this being a modelling situation. Relatively fewer candidates were successful in part (b)(ii). Some who had realised they were to solve for t , with x set to zero, did not deal with the 2 in the expression before exponentiation. A few candidates, who were otherwise correct, overlooked taking the square root in finding t .

Q9.

Not all candidates appeared to have time to attempt this question and often solutions appeared rushed rather than thought out. However many of those who did attempt the question solved the differential equation in part (a) correctly. Most candidates separated the variables correctly using conventional notation, but then there were often errors in the integration. Common ones included the omission of a constant, incorrect integration of both x and $\cos 2t$ or use of degrees instead of radians when evaluating the constant. Some candidates just substituted the given conditions straight into the differential equation, producing nonsense.

In part (b)(i) some candidates substituted $\pi/4$ into dx/dt instead of their solution from part (a). Another error was to substitute $13\pi/4$ for t . There were, however, some correct solutions showing understanding and most candidates who attempted this did show they intended to work with $\sin 26$, even if they were trying to use it in an incorrect expression or with their calculator in degrees. Most gave answers to the accuracy required by the

question.

In part (b)(ii) those who had successfully solved the differential equation in part (a) usually earned the method mark here. However correct answers were rare, as many candidates arrived correctly at a negative value for $2t$, and just ignored the negative sign, rather than considering the next positive solution. Another common error lay in using the wrong calculator mode once more.

Q10.

The responses to this question were very mixed, varying from the very poor to some excellent answers. Most, but not all, candidates did attempt to separate and integrate in part (a)(i) with some making an error in their integrals, and rather more omitting a constant. Such candidates could make no further progress in part (a)(ii) without a constant to find. Some candidates thought to bring e^{-1} into their answer in part (a)(i), instead of the expected conventional constant, and got themselves confused in part (a)(ii) as they could not convincingly obtain the given answer. Some candidates made the anticipated error at the exponentiation stage of $\ln y = -\cos t + c \Rightarrow y = e^{-\cos t} + e^c$. Many candidates could not justify the 50, other than saying it was the initial value.

For part (b) many candidates realised they had been given the required expression in part (a) and substituted and evaluated correctly, although some ignored the request to answer to the nearest centimetre, any greater accuracy being rather meaningless in this modelling problem. Some used degrees instead of radians. Other candidates substituted into the differential equation. In part (b)(ii), although most candidates knew in principle what to do, they could not find the second derivative correctly. Many treated y as a constant or as if it were t , rather than using implicit differentiation.

Some candidates did attempt to differentiate the given answer in part (a)(ii) twice, but usually made an error in their use of the chain rule. Some candidates did get the differentiation correct, but failed to complete their argument by demonstrating that the first derivative was zero at $t = \pi$, so confirming this point as a turning point.

Q11.

Most knew in principle that they were to separate the variables and integrate, but there

was a lot of mishandling of the $(x - 6)^{\frac{1}{2}}$ term. Most candidates integrated dt correctly, and included a constant, and so were able to show they could use the initial conditions to find a value for the constant even if their x term was incorrect. Common errors in the integration of the x term were in the coefficient of $(x - 6)^{\frac{1}{2}}$ or multiples of $(x - 6)^{\frac{3}{2}}$ or $\ln(x - 6)^{\frac{1}{2}}$.

Some candidates who had done all the integration correctly failed to give their final answer in the form requested.

For part (b) most candidates showed they understood the context through making a sensible comment in part (i), although a few thought it meant the tank was empty.

Those who just stated $\frac{dx}{dt} = 0$ gained no credit.

In part (b) (ii) quite a few candidates lost the marks because they used $x = 48$ ($70-22$) rather than $x = 22$. However, there were many candidates who gave good full answers to this question even if a few lost sight of the correct units for the final answer. This was condoned.

Q12.

Many candidates gained full marks on this question presenting clear, good quality answers.

Part (b)(i) Most candidates knew they were required to separate and integrate and attempted to do so, sometimes with poor notation and incorrect integrals but their intention was clear. Most candidates also included a constant of integration and attempted to find it using the given initial conditions. Some neglected to express t in terms of x with some going straight to the result given in part (b)(ii).

Part (b)(ii) This was often done well but the common error was in taking exponentials, with $e^{\frac{t}{40}} = e^{\ln 70} + e^{\ln(x-15)}$ being fairly common.

Q13.

This question was usually answered well. However some students forgot the minus sign in

the initial equation $12 \frac{dv}{dt} = -4v^{\frac{1}{3}}$ whilst others had difficulty with the evaluation of $\int v^{-\frac{1}{3}} dv$.

Most students attempted to find the '+c' term and obtained the printed result. Nearly all students correctly found the time when the particle came to rest.

Q14.

Most students answered part (a) correctly but some did not replace their a by $\frac{dv}{dt}$. In part

(b) students were expected to use $\int \frac{1}{100-v} dv = -\int \frac{1}{40} dt$; some incorrectly split $\int \frac{1}{100-v} dv$ into $\int \frac{1}{100} - \frac{1}{v} dv$, whilst others incorrectly used $\int (100-v) dv = \int \frac{1}{40} dt$.

A common error was in forgetting the minus sign in the evaluation of $\int \frac{1}{100-v} dv$. Most students attempted to find the '+ c' term in the solution of the differential equation.

Q15.

This question was answered well by most candidates. However, in part (a), a few

candidates did not show the required step $0.4 \frac{dv}{dt} = 2 - 4v$; others lost and recreated the

minus signs, eg $\frac{dv}{dt} = 5 - 10v = -10(0.5 + v) = -10(v - 0.5)$, which was penalised. In part (b), the majority of candidates used $\int \frac{1}{v-0.5} dv = - \int 10dt$ and then solved this by integrating both sides.

The integration $\int \frac{1}{v-0.5} dv$ caused some candidates difficulty, finding terms such as $\frac{1}{\frac{1}{2}v^2 - 0.5v}$ etc. The use of logarithms was often incorrect, with $\ln(v - 0.5) = -10t + \ln 0.5$ commonly becoming $v - 0.5 = e - 10t + 0.5$

Q16.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although some used inventive algebra. The solution to part (c) was usually correct, although candidates often forgot to obtain ± 1 , for the square root of v , when they found the square-root of the equation obtained in part (b).

In part (d) candidates who integrated v to find the distance travelled often forgot the $+d$ term which was required to be shown to be equal to zero.

Those who used integration with limits commonly used $\int_{16}^1 (4 - 0.1t)^2 dt$ forgetting that the integral was with respect to t so that $\int_0^{30} (4 - 0.1t)^2 dt$ was required. A considerable number of candidates, in part (d), tried to use equations for motion with constant acceleration.

Q17.

The answer printed in part (a) enabled virtually all candidates to create their solution. However, a considerable minority were not convincing. Part (b) was well answered, with most candidates appreciating that the differential equation needed to be solved by

separating variables. Some found the resulting integral, $\int v^{-\frac{3}{2}} dv$, challenging, and the $+c$ term was often omitted. In part (c), rather than using $\frac{2}{\sqrt{v}} = 3t + \frac{2}{3}$ to obtain t when

$v = 4$, many used the last result, $v = \frac{36}{(2 + 3t)^2}$ when $v = 4$, and forgot to eliminate the negative root when calculating the square root of $4 = \frac{36}{(2 + 3t)^2}$.

Q18.

Parts (a), (b) and (c) were answered well. In part (d), few candidates used the fact that the work done by the force was equal to the change in kinetic energy. Most used complicated methods which rarely gave an appropriate answer.

Q19.

Virtually all candidates obtained $\frac{dv}{dt} = -0.05v$; only a few ignored the required step $m \frac{dv}{dt} = -0.05mv$. The equation $\int \frac{dv}{v} = - \int 0.05 dt$ was a necessary step which needed to be seen in part (b). Candidates knew roughly how to obtain $v = 20e^{-0.05t}$. Unfortunately, too often algebraic skills were not sufficient and the equation $\ln v = -0.05t + c$ regularly became $v = e^{-0.05t + c}$ before becoming $v = Ae^{-0.05t}$. This and similar errors were not condoned.

Q20.

Part (a) of this question was often done well. Many candidates seemed to take the question in their stride, but others made limited progress or did not know how to approach the question. There were some who lost accuracy marks. The four main causes of this were;

- Omission of the negative sign in the differential equation,
- Algebraic errors,
- Incorrect integration,
- Not finding the value of the constant of integration.

Part (b) was done very well, with many candidates gaining the available mark without doing part (a).

Q22.

There were a number of good solutions to part (a). The main errors were due to missing negative signs in the differential equation, the use of $v \frac{dv}{dx}$ instead of $\frac{dv}{dt}$ or algebraic errors in the working. Very few candidates tried to use the constant acceleration equations. Part (b) was answered correctly by the vast majority of the candidates.